Titanium Ti-6Al-4V ELI (Grade 23), Annealed

Categories: Metal; Nonferrous Metal; Titanium Alloy; Alpha/Beta Titanium Alloy

Material Notes: Information provided by Allvac and the references. Annealing Temperature 700-785ºC. ELI (Extra Low Interstitial) grade has lower impurity limits, especially oxygen and iron. Alpha-Beta Alloy

Applications: Applications requiring excellent fracture toughness and fatigue strength; aircraft, structural components, and biomedical implants.

Biocompatibility: Excellent, especially when direct contact with tissue or bone is required. Ti-6Al-4V's poor shear strength makes it undesirable for bone screws or plates. It also has poor surface wear properties and tends to seize when in sliding contact with itself and other metals. Surface treatments such as nitriding and oxidizing can improve the surface wear properties.

4 other heat treatments of this alloy are listed in MatWeb.

Key Words: Ti-6-4; ASTM Grade 23 titanium; UNS R56401 (ELI); Ti6Al4V, Ti64, biomaterials, biomedical implants, biocompatibility

Vendors: No vendors are listed for this material. Please click here if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	**Metric**	**English**	**Comments**
Density	4.43 g/cc	0.160 lb/in³	

Mechanical Properties	**Metric**	**English**	**Comments**
Hardness, Brinell	326	326	Estimated from Rockwell C.
Hardness, Knoop	354	354	Estimated from Rockwell C.
Hardness, Rockwell C	35	35	
Hardness, Vickers	341	341	Estimated from Rockwell C.
Tensile Strength, Ultimate	860 MPa	125000 psi	
Tensile Strength, Yield	790 MPa	115000 psi	
Elongation at Break	15 %	15 %	
Modulus of Elasticity	113.8 GPa	16510 ksi	
Compressive Yield Strength	860 MPa	125000 psi	
Notched Tensile Strength	1170 MPa	170000 psi	K_t (stress concentration factor) = 3.5
Ultimate Bearing Strength	1740 MPa	252000 psi	e/D = 2
Bearing Yield Strength	1430 MPa	207000 psi	e/D = 2
Poissons Ratio	0.342	0.342	
Fatigue Strength	140 MPa @# of Cycles 1.00e+7	20300 psi @# of Cycles 1.00e+7	K_t (stress concentration factor) = 3.1
	300 MPa @# of Cycles 1.00e+7	43500 psi @# of Cycles 1.00e+7	unnotched
Fracture Toughness	100 MPa-m½	91.0 ksi-in½	K_{IC}
Shear Modulus	44.0 GPa	6380 ksi	
Shear Strength	550 MPa	79800 psi	Ultimate shear strength
Charpy Impact	24.0 J	17.7 ft-lb	V-notch

Electrical Properties	**Metric**	**English**	**Comments**
Electrical Resistivity	0.000178 ohm-cm	0.000178 ohm-cm	
Magnetic Permeability	1.00005	1.00005	at 1.6 kA/m
Magnetic Susceptibility	0.0000033	0.0000033	cgs/g

Thermal Properties	**Metric**	**English**	**Comments**

CTE, linear	8.60 µm/m-°C @Temperature 20.0 - 100 °C	4.78 µin/in-°F @Temperature 68.0 - 212 °F	
	9.20 µm/m-°C @Temperature 20.0 - 315 °C	5.11 µin/in-°F @Temperature 68.0 - 599 °F	average
	9.70 µm/m-°C @Temperature 20.0 - 650 °C	5.39 µin/in-°F @Temperature 68.0 - 1200 °F	average
Specific Heat Capacity	0.5263 J/g-°C	0.1258 BTU/lb-°F	
Thermal Conductivity	6.70 W/m-K	46.5 BTU-in/hr-ft²-°F	
Melting Point	1604 - 1660 °C	2919 - 3020 °F	
Solidus	1604 °C	2919 °F	
Liquidus	1660 °C	3020 °F	

Component Elements Properties	Metric	English	Comments
Aluminum, Al	5.5 - 6.5 %	5.5 - 6.5 %	
Carbon, C	<= 0.080 %	<= 0.080 %	
Hydrogen, H	<= 0.0125 %	<= 0.0125 %	
Iron, Fe	<= 0.25 %	<= 0.25 %	
Nitrogen, N	<= 0.030 %	<= 0.030 %	
Other, each	<= 0.10 %	<= 0.10 %	
Other, total	<= 0.40 %	<= 0.40 %	
Oxygen, O	<= 0.13 %	<= 0.13 %	
Titanium, Ti	88.1 - 91 %	88.1 - 91 %	As Balance; Elemental Composition per ASTM B265
Vanadium, V	3.5 - 4.5 %	3.5 - 4.5 %	

Descriptive Properties			
Velocity of Sound		4.987 km/s	Heat treatment not specified

References for this datasheet.